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Hospital Patient Records Cleaning and Health Metrics Dashboard

Project Domain

Power BI with Copilot

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:

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3. Problem Statement

Hospitals generate a large amount of patient data every day, including patient demographics, diagnoses, treatment records, admission and discharge dates, and health indicators. However, hospital datasets are often **messy, incomplete, duplicated, or inconsistent**, which makes it difficult for healthcare administrators to extract meaningful insights.

Unclean data may contain **missing patient IDs, incorrect age values, duplicate records, inconsistent gender labels, and incorrect date formats**. Such issues reduce the reliability of data analysis and can affect decision-making in healthcare management.

Healthcare professionals and hospital administrators rely on accurate data to monitor **patient admissions, disease trends, hospital capacity, treatment outcomes, and overall healthcare performance**. Without proper data cleaning and visualization, it becomes difficult to track health metrics or identify patterns that could improve hospital services.

This project focuses on **cleaning hospital patient records and building a Health Metrics Dashboard** using tools such as **Excel and Power BI**. The project will demonstrate how raw healthcare data can be transformed into clean, structured datasets and visual dashboards that support better decision-making.

The solution will help stakeholders such as **hospital administrators, doctors, and healthcare analysts** understand key metrics like patient admissions, disease distribution, average length of stay, and recovery rates.

4. Objectives

The main objectives of this project are:

• To collect and explore a dataset containing hospital patient records and understand its structure and data quality issues.

• To clean and preprocess the dataset by removing duplicate entries, handling missing values, correcting data formats, and standardizing fields such as gender, age, and diagnosis categories.

• To perform data analysis using Excel formulas, pivot tables, or Power BI measures to identify important healthcare trends and statistics.

• To design an **interactive Health Metrics Dashboard** that visually represents hospital data through charts, graphs, and key performance indicators (KPIs).

• To integrate AI tools such as **Microsoft Copilot or ChatGPT** to assist in data interpretation, report writing, and generating analytical insights.

• To document the entire workflow including data cleaning steps, dashboard development, and AI integration.

• To provide actionable insights that can help hospital administrators improve patient care and operational efficiency.

5. Methodology / Implementation Plan

The project will follow a structured five-phase workflow:

Phase 1: Data Collection & Exploration

A hospital patient dataset will be obtained from sources such as **Kaggle or other open healthcare datasets**. The dataset will include fields such as patient ID, age, gender, diagnosis, admission date, discharge date, treatment type, and hospital department. Initial exploration will be performed to understand the dataset structure and identify inconsistencies.

Phase 2: Data Cleaning & Preprocessing

Data cleaning will be performed using **Excel or Power Query**. This phase will include removing duplicate patient records, filling or removing missing values, correcting inconsistent formats (e.g., date formats), and standardizing categories such as gender or diagnosis labels.

Phase 3: Analysis & AI Integration

Data analysis will be conducted using **Excel pivot tables, formulas (SUMIF, COUNTIF, VLOOKUP)**, or **Power BI DAX measures**. AI tools such as **ChatGPT or Microsoft Copilot** will be used to generate summaries of the analysis and assist in interpreting healthcare metrics.

Phase 4: Visualization / Dashboard Creation

An interactive **Health Metrics Dashboard** will be developed using **Power BI or Excel charts**. The dashboard will display important indicators such as:

- Total number of patients
- Patient admissions by department
- Disease distribution
- Average length of hospital stay
- Gender distribution
- Monthly patient admission trends

Phase 5: Documentation & Submission

All project work will be documented in a structured report. The cleaned dataset, dashboard files, and documentation will be uploaded to a **GitHub repository**, and a **video demonstration** of the dashboard will be recorded and submitted.

6. Resources and Timeline

A. Hardware & Software Requirements

• Laptop or PC (Intel Core i3 or higher, minimum 4 GB RAM)
• Operating System: Windows 10 / 11 or macOS
• Microsoft Excel (2019 or Microsoft 365)
• Power BI Desktop
• Microsoft Copilot
• Stable Internet Connection
• Web browser (Google Chrome or Microsoft Edge)

B. Dataset

The dataset for this project will consist of **hospital patient records** containing demographic and treatment-related information.

Possible sources include:

• Kaggle Healthcare Datasets
• UCI Machine Learning Repository
• Simulated hospital datasets if real data is unavailable

Example dataset fields:

- Patient ID
- Age
- Gender
- Diagnosis
- Admission Date
- Discharge Date
- Department
- Treatment Outcome

C. Project Timeline

Phase	Task	Tentative Completion
1	Dataset collection and exploration	Week 1
2	Data cleaning and preprocessing	Week 2
3	Data analysis and AI integration	Week 2–3
4	Dashboard design and testing	Week 3
5	Report writing and project submission	Week 4

7. Expected Outcomes / Conclusion

By the end of the project, the following outcomes are expected:

• A **clean and structured hospital patient dataset** with documented preprocessing steps.

• An **interactive Health Metrics Dashboard** that provides visual insights into hospital operations and patient statistics.

• Identification of important healthcare trends such as disease prevalence, patient demographics, and hospital workload distribution.

• Demonstrated use of **AI tools** for data interpretation and report generation.

• A **GitHub repository** containing the dataset, dashboard files, documentation, and project resources.

• A video demonstration explaining the project workflow and insights derived from the dashboard.

This project will demonstrate how **data cleaning, analysis, and visualization combined with AI tools can improve healthcare data management and support better decision-making in hospitals.**

8. References

- Edunet Foundation. *AICW Fellowship Program Guidelines.* 2026.
- Microsoft Corporation. *Microsoft Excel Documentation.*
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- LinkedIn Learning. *Data Analytics and AI Courses.*
- Microsoft Copilot Documentation.